PROJECT QUALITY CONTROL

(Doc No. CT8-PRO-CM-001)

USA PROJECT UNIT 1 - TALLGRASS

Revision History				
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Signature:					

BJCHI Heavy Industries Public Company Limited

VISION MISSION BUSINESS GOAL
QUALITY ENVIRONMENTAL AND OCCUPATIONAL HEALTH AND SAFETY

VISION

To be a world-class integrated engineering, procure-ment, fabrication and modularization service provider

MISSION

- To create sustained value for our shareholders.
- To meet customers' needs through exceptional quality and on-time delivery.
- To continuously improve to ensure operational excellence and competitive edge.
- To develop our people.
- To contribute to society and protect the environment.

BUSINESS GOAL

BJCHI aims to achieve long-term sustainable growth, while balancing the interests of all the stakeholders. To this end, we are striving to:

- Deepen and diversify our Portfolio, building on our track record of successful project deliveries;
- Preserve our Profitability by improving productivity, enhancing safety and reducing costs;
- Enhance performance through simplification, standardization and integration of Processes;
- Develop People to underpin our growth.

QUALITY ENVIRONMENTAL AND OCCUPATIONAL HEALTH AND SAFETY POLICY

1. Produce goods of quality under international Standardization and safety in competitive prices
2. On-time delivery effectively to meet the needs of the customer
3. .Committed to practice compliance obligations that must be followed. Legal and other requirements related to the organization in quality environmental and occupational health and safety.
4. Committed continual improvement to the company standard system to create customer satisfaction and stakeholders
5. Commit proactive in develop of employees capacity and create Consciousness in work in quality, environmental and occupational health and safety.
6. Committed communicate and promote the consultation, participation and awareness of personnel and stakeholders in terms of quality, environmental, occupational health and safety.
7. Committed allocate provide resources needed in develop of employees capacity in quality, environmental and occupational health and safety.
8. Committed to provide a safe working environment Hygienic to prevent injury and work-related illness. Property damage Environmental damage and working environment area or the public
9. Committed reducing the risk of accidents in high level that will impact personnel in the organization and stakeholders to stay accepted criterion including continuous improvement
10. Committed prevent air pollution noise and wastewater affecting surrounding communities and stakeholders
11. Committed protect environmental resources and safety in working of personnel
12. Commit proactive in develop the organizational capacity of the organization in quality environmental and occupational health and safety.
13. Committed to promote awareness among the personnel into reduce waste, reduce energy use including resource usage
14. Determine objectives and quantitative targets for improve and ongoing operations the in quality, environmental and occupational health and safety

Mr. Young Jun Lee
Managing Director
BJC Heavy Industries Public Company Limited
Date 15 March 2024

Number : VMP (Eng) - BJCHI-001 Rev.2
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ORIGINAL CONTROLLED

CONTENTS

1. SCOPE 4

2. APPLICATION OF THE PQP 4

3. NORMATIVE REFERENCE AND DEFINITIONS 4

4. PROJECT QUALITY MANAGEMENT SYSTEM 5

5. MANAGEMENT RESPONSIBILITY 6

6. RESOURCE MANAGEMENT 11

7. PROJECT REALIZATION 12

8. MEASUREMENT, ANALYSIS AND IMPROVEMENT 21

1. SCOPE

This Project Quality Plan ("PQP") describes how BJCHI will apply our Quality Management System, which has been established in accordance with the requirements of the International Standardization (ISO 9001;2015), to meet the specific requirements.

This PQP establish the Project Quality System and describes the detailed procedural requirements to assure the reliability in quality of products and services for the project in compliance with the contract.

The particular procedures of the Project Quality System which are applicable to each discipline of this Project organization are referred to each section of this PQP.

2. APPLICATION OF THE PQP

The Quality Assurance & Quality Control Manager ("QAQCM") is responsible for the preparation, issue and distribution of the PQP, including subsequent revisions.

The PQP shall apply to the activities of CONTRACTOR to ensure that all works supplies and services for the Project are carried out in a planned and systematic manner based on the guidelines of this PQP.

The PQP and supporting control documents shall be under constant review and shall be amended, added to or deleted form, as is appropriate, following the experience gained as the Project progresses.

3. NORMATIVE REFERENCE AND DEFINITIONS

ISO 9000 :	Quality Management Systems - Fundamentals and vocabulary
ISO 9001 :	Quality Management Systems - Requirements
Owner :	Project owner or Owner's representative
Project :	USA Project Unit 1 - Tallgrass
Product :	end result of a process.
Process :	set of interrelated or interacting activities which transforms inputs into outputs
ITP :	Inspection and Test Plan for site and shop inspection control
Procedure :	specified way to carry out an activity or a process
Quality Management System (QMS):	management system to direct and control an organization with regard to quality
Quality Plan:	Document specifying which procedures and associated resources shall be applied by whom and when to a specific project, product, process or contract
QAQC	Quality Assurance & Quality Control
HSE	Health, Safety and Environment
PM	Project Manager
QM	Quality Manager
EM	Engineering Manager
MM	Modularization Manager
PPM	Project Procurement Manager
WHM	Warehouse Manager

Note: Terms and definitions used are consistent with the definitions of ISO 9000 standard, unless otherwise specified in a different way in the Project Quality Plan.

4. PROJECT QUALITY MANAGEMENT SYSTEMS

4.1 Introduction of Project Quality Plan

This project quality plan establishes the applicable quality assurance program requirements, identifies or references detailed implementing procedures, practices, and plans for the Project, including engineering, procurement, construction and commissioning activities to be performed by CONTRACTOR.

For completeness, and for developing awareness of the project task force, all of the requirements from ISO 9001 standard are listed.

The quality system, as described in this project quality plan, is based on the following fundamental principles adhered to by CONTRACTOR.

The achievement of quality is a line responsibility wherein each performer and supervisor are accountable for the quality of the work assigned to. This applies to all personnel responsible for engineering, procurement, construction and commissioning.

The quality organization maintains a strong overview of work to give additional assurance that specified requirements are met. This is done by performing reviews and approvals of documents; by conducting inspection, surveillance and quality audit.

The review and approval of quality documents is at a management level commensurate with the importance of the item under consideration.

It is intended that this document will supplement, rather than replace, CONTRACTOR's existing ISO 9001 certified quality management manual and system procedures.

Where necessary, existing proven procedures will be tailored to meet the needs of the Project to implement all services in an efficient and effective manner. All project task-force personnel will work under this quality program as approved by OWNER and CONTRACTOR management.

The project quality management principles are supported by approved, proven, and controlled quality manuals, plans, procedures, work instructions, checklists, etc.

4.2 Documents Control

CONTRACTOR controls all applicable internal and external documents and data such as Quality Procedures, Standards/Codes, Drawings, Inspection & Test Plans, and Product Specifications etc. per Document and Distribution Procedure, Document Review and Approval Procedure, which is defined the conditions needed for document identification, review and approval, distribution, changes and obsolete documents control.

CONTRACTOR maintains the Master Document Index to help an organization keep track of documents currently available. All documents in the Master Document Index contain effective date and current revision level.

CONTRACTOR also ensures only pertinent issues of appropriate documents are available for use and obsolete documents are promptly removed and suitably identified to prevent inadvertent use.

Assigned document controllers are responsible for handling the engineering documents, procurement documents, (including subcontractors and vendors documents) etc., such as receiving, listing, filing, delivering and distribution of documents on daily basis.

Documents shall be prepared by each discipline shall be checked and approved by authorized person. Document controllers shall send specified number of copies to the pertinent location with a transmittal sheet on each copy in accordance with Document and Distribution Procedure.

4.3 Records Control

Project records as objective evidence will be identified and maintained to demonstrate that the specified quality and progress has been achieved. These records will include, but not be limited to, all records related to management, control, engineering, procurement, construction, mechanical completion and commissioning activities, as specified throughout this project quality plan and implementing manuals, plans, and procedures, and in conformance with applicable requirements of the Contract.

Project records will include results and evidence of quality verification activities such as inspections, tests, checking, surveillance and audit reports, NCRs, welding procedure specifications, quality procedures used, quality certificates of conformance, etc.

CONTRACTOR maintains records per relevant records control procedures, which establishes a standard method for identification, collection, storage, protection, retrieval, retention time and disposition of records.

The records absolutely must be accurate, legible, retrievable and stored in a permanent medium, if possible.

The specific project records control system is applied to the project/contracts which CONTRACTOR agrees the specific requirements of OWNER. The project records control system shall be established on the basis of the contract requirements of the project.

5. MANAGEMENT RESPONSIBILITY

5.1 Management Commitment

PM shall ensure establishing, implementation, evaluating of the quality management system and continual improvement of its effectiveness, and also ensure that the organization realizes the customer satisfaction by

- To suggest directions in order to satisfy statutory, regulatory requirement and OWNER requirements,
- To ensure that spreading quality policy, establishment of quality objectives consistent with the quality policy and establishment of implementation plan to achieve quality objectives,
- To ensure that continual review and improvement of the quality management system for customer satisfaction through management review on a regular basis,
- To ensure that continual improvement of processes, obtaining the necessary management resources for the work and its effective operation.

5.2 Customer Focus

PM ensures that work processes shall be identified at an appropriate time to meet OWNER requirements and expectations including statutory and regulatory requirements regarding products and services to be provided by CONTRACTOR. And also ensures that the criteria and methods to reflect them are determined and implemented for customer satisfaction.

5.3 Planning

5.3.1 Project Quality Objectives

CONTRACTOR is responsible for engineering, procurement, construction and commissioning activities of the Project, as required by the Contract. It is the general project quality objectives to deliver all products and services required under the terms and conditions of the Contract, in full compliance with the project specifications, OWNER requirements, and all relevant statutory requirements.

5.3.2 Project Quality Management System Planning

The project quality system includes the followings:

The first level consists of the quality management planning, which includes this project quality plan, the project execution plan, Field quality control plan (Inspection & Test Plan), and the project procedures, etc. for Home Office and Site, which will be used by the project team.

The second level consists of the quality control checks and reviews, such as those described in the engineering, procurement, construction, and commissioning procedures, and discipline activity plans, work instructions, and checklists that will be used by the project team.

The third level consists of the inspections, including witnessing of inspections, tests and other activities, quality audits and surveillance, reviews by quality personnel.

PM will ensure that the first-level documents are distributed to the appropriate personnel as soon as they are approved for issue, and that all updates or revisions are similarly distributed during the project life.

The second and third level documents are prepared by the appropriate discipline and quality personnel and, after being verified and approved by the discipline and project management, are distributed as required.

The PM should be responsible for ensuring that the 2nd and 3rd level documents are issued and in place in time for the work to proceed.

5.3.3 Project Quality System Review

Project quality system will be reviewed by QM to evaluate project performance in respect of project quality requirements. QM will ensure that the quality plan and procedures are reviewed on a regular basis to verify their continued applicability and those they accurately reflect the intended purpose.

In addition, these reviews will address recorded non-conformances with a view to identifying any trends that could be adverse to quality and to determine any timely corrective actions required to ensure compliance with the project quality requirements.

Any concerns raised will be recorded and will include details of the personnel designated to undertake corrective actions where required.

5.3.4 Responsibility, Authority and Communication

1) Organization

At the project level, the project organization consists of project management, project quality, project HSE, engineering, procurement, construction and commissioning. Manager of each function shall direct report to project manager.

The overall project execution organization and the structure of the various sub-organizations, showing key personnel, lines of communication, and interfaces, are shown in Project Coordination Procedure. And typical Project Quality Organization Chart is shown in ***Attachment#1*** of this PQP.

2) Responsibilities of Project Key Personnel

- **Project Manager (PM) has the following responsibilities;**
  - Plan, organize, direct, execute and coordinate the project.
  - Supervise, direct and control all personnel assigned to the project.
  - Be responsible for execution of the contract to obtain anticipated job quality, effective cost control, and compliance with job schedule.
  - Be responsible to Management and the client for the project.
  - Define project scope and services. And issue project instruction.
  - Review and/or prepare the contract document.
  - Prepare change orders as required.
  - Be responsible for issuing the project budget and the project cost reports.
  - Be responsible, through Engineering Manager, for design and engineering work.
  - Be responsible, through Procurement Manager, for selecting vendors and approving bid evaluation documents for purchase order.
  - Be responsible to the Modularization Manager for performance be responsible, Construction Manager for construction.
  - Be responsible, through Commissioning Manager, for commissioning & start-up, initial operation and performance test run (if any).
  - Obtain Client's acceptance of the plant and arrange for close out of the project
  - Initiate kick-off meetings and project staff meetings.
  - Report to project management and Client.
  - Approve and issue all project documents.
  - Be responsible the project to meet the criteria of effective cost control and compliance with project schedule.
  - Follow up with responsible parties to ensure that schedules are maintained or improved.
  - Assure the scheduling and/or cost control report by planning, analyzing, and forecasting of project performance for management.
  - Assess the effects of all deviations for project schedule, at the earliest possible time, for evaluation and remedial action by Project Manager.
  - Preparation of all trend, status and progress reports.
  - Assure communications between other engineering and construction disciplines.

- Be responsible to the Project Manager for controlling and handling all of project documents and keeping all originals of studies, specifications, drawings, sketches, requisitions, calculation and data sheets, vendor prints etc.
- Document filing control.

➢ **Quality Manager (QM/PQM) has the following responsibilities,**

- Assist and coordinate with Project Manager and all personnel assigned to the project in establishment of Project Quality Plan.
- Be responsible for ensuring that project quality management procedures are implemented effectively and are suitable for the purposes and use for which they are intended.
- Establish project quality audit program & schedule and perform the project quality audit within project organization.
- Monitoring & Evaluation of quality management system activities based on the project quality plan.
- Coordination with Client for project quality management system concerns.
- Act as the focal point for all site quality activities and shall maintain close liaison with the client and/or the appointed agencies.
- Provide technical assistance to site management as necessary to initiate quality enhancement and resolve quality problems throughout the organization.
- Monitor, provide technical assistance and resolve all quality-related problems with the activities of subcontractors.
- Prove in accordance with the applicable quality and inspection requirements imposed by the contract the qualification and experience of all inspection personnel and monitor and audit on frequent basis to ensure technical compliance.
- Plan for and implement the quality training programs at site.
- Establish in accordance with contractual requirements an Inspection & Test Plan (ITP) indicating the extent of inspection based on item criticality.
- Ensure that adequate inspection personnel and test tool & equipment resources are available at all times, including peak load periods to meet the construction schedule.
- Raise non-conformance items, investigate their causes and make recommendations for corrective actions to prevent recurrence.
- Ensure all technical documentation is accurate and updated to current status.
- Conduct the education for quality consciousness to each subcontractor prior to the commencement of work.
- Undertake quality audit to all subcontractors employed on site on a frequent basis to ensure quality compliance.
- Give his staffs the instruction about duties and responsibilities on site.
- Supervise his staffs' activities, functions and operations so that they are taken in accordance with the specified requirements and the approved procedures.

- **Engineering Manager (EM) has the following responsibilities.**
  - Be responsible to the Project Manager for planning, coordinating and directing the engineering and design work to meet the criteria of anticipated job quality with economic design.
  - Each EM has responsible for monitoring and implementing of design QC activity each engineering party within his own work scope.
  - Supervise and coordinate the design work of the various engineering disciplines.
  - Establish engineering and design basis. And also define the deliveries and non-deliveries.
  - Work with the lead engineers to formulate, evaluate and execute practical, economical and efficient design concepts suitable for the specific requirements of the project.
  - Resolve the conflicts among the engineering disciplines during the engineering and design work.
  - Monitor engineering and design progress and report problems to the Project Manager for resolution.
  - Chair internal engineering discipline review meetings and prepare the notes of each meeting.
  - Alert the Project Manager of any potential design, schedule and/or cost impact of design work or changes.
  - Review and approve as delegated, key engineering documents such as requisitions, bid summaries, material specifications, engineering subcontracts, P & ID, plot plans, etc. before they handed over to Project Manager, to assure conformance with the project requirements.
  - Coordinate with Design Safety LE for engineering HSE concern.
- **Project Procurement Manager (PPM) has to following responsibilities.**
  - Be responsible to the Project Manager for the coordination of all procurement activities between engineering, construction, purchasing, expediting, forwarding, traffic, accounting, law, insurance and tax on his assigned project.
  - Maintain contact with his counterparts in the client's organization.
  - Represent the home office procurement team within the project. Attend and contribute the project staff meetings as necessary.
  - Keep the Project Manager informed of procurement status in general and matters important or critical to the project.
  - Prepare, issue and revise procurement procedures.
  - Supply guidance and counsel to project personnel as regards procurement procedures, content and format of basic documents such as Requests for Quotation (RFQ), Quotation Summaries, Purchase Orders, etc.
  - Act as liaison between project engineering and the procurement team on all requests for and approvals of, physical expediting. Keep informed as to progress and results of the expediting effort.

- **Modularization Manager (MM) has the following responsibilities.**
  - Be responsible for coordinating overall execution at site.
  - Set up and administer general operation of field office including camp facilities.
  - Review constructability thoroughly prior to site opening. Coordinate with the construction members to prevent or solve possible problems.
  - Supervise and control field staffs in site organization. Delegate and assign to each group of site organization all necessary functions to be performed during the phase of construction, commissioning & start-up. Check construction-planning documents to see whether any problem may arise in future. Plan for preventive measure or set up a contingency plan.
  - Establish a plan of construction execution considering actual site conditions.
  - Direct field purchasing activities including material warehousing, and expedite site delivery.
  - Preside weekly schedule meeting, weekly safety meeting, and monthly progress review meeting to assure that all activities are performed in accordance with original planning.
  - Prepare construction status reports for the client and home office.
  - Direct all construction activities under execution and evaluate the result of field test.
  - Direct site HSE performance in determining safety measures to be taken by Site HSE Manager. Keep Project Manager informed of the result of accident handling.
  - Approve requests for field extra work order, change order, and memorandum work.
  - Maintain cooperative contact with client's project manager.
  - Submit monthly progress report to the client and expedite the approval on it.
  - Submit monthly invoice for erection portion to the client and expedite the payment of it.
  - Approve various field-originated reports to be submitted to the home office.
  - Prepare Final Construction Report.

3) Internal Communication

Project internal communication system shall be established and implemented inside and outside project team in order to achieve quality policy and quality objectives and to obtain effectiveness of quality management system and work processes.

PM shall be aware of the importance of internal communication and ensure communication for the establishment of work processes and the effect of quality management system.

6. RESOURCE MANAGEMENT

6.1 Provision of Resources

CONTRACTOR shall identify, determine and mobilize resources appropriate to provide products and services required by OWNER, and allocate/assign the mobilized resources in a timely manner. The mobilized resources shall ensure that implementation, maintenance of quality management system and continual improvement of its effectiveness.

6.2 Human Resources

The goal for human resources is always hire competent personnel, who are performing work affecting product quality.

PM will ensure that adequate human resources with the necessary expertise are available to enable the efficient execution of the Project. In this respect, he will ensure that any required specialist skill not available within the organization is identified at an early stage and appropriate arrangements are made to engage qualified personnel or subcontractors with relevant experience.

Personnel competencies are based on appropriate education, training, skills and experience in accordance with the applicable CONTRACTOR Management Standards

Each department/team manager ensures that the personnel within his scope are aware of the relevance and importance of their activities and how they contribute to the achievement of the quality objectives.

6.3 Infrastructures and Work Environment

Project management determines, provides and maintains the infrastructure necessary for project execution while considering contractual requirements and project internal needs.

The infrastructures include, as applicable, the follows

- Buildings, workspace and associated utilities
- Process equipment (both hardware and software)
- Supporting Services such as transport or communication

The infrastructures needed are provided and verified at the stage of project execution planning or prior to starting the work.

Project management determine and manage the work environment needed to achieve conformity to product requirements.

The work environment conditions include physical, social, psychological and environmental factors such as temperature, recognition schemes, ergonomics and atmospheric composition.

7. PROJECT REALIZATION

7.1 Planning of Project

7.1.1 Kick-off

PM has responsibility to establish the Project Execution Plan in accordance with project specification as soon as the project is awarded.

PM shall call an internal kick-off meeting as earlier as possible, to which all relevant Construction Manager (MM) and discipline lead engineers (LE) shall attend in reference of following project specification. During the kick-off meeting, project execution plan including major project highlights, master schedule and contents of job instructions are explained for overall understanding of the project, and each manager is requested to establish detailed activity plan.

7.1.2 Contract Review (Internal Use Only)

At the receipt of contract package, CONTRACTOR shall conduct preliminary review to evaluate the plant in technical view on safety, design and fabrication maturity, and also in contractual view on cost and schedule, and to ensure the adequacy of requirements, information, data and the capability to meet contractual requirements.

When any incomplete, ambiguous and conflicting items are found, these items shall be clarified by written document or official meeting.

Based on contract review, PM will prepare job instruction that contains necessary information to start job, and draw up Project Execution Plan.

7.1.3 Quality Activity Plan

As part of overall planning and scheduling for project realization, project organization shall develop project management tools to provide a clear, complete and consistent definition of the following.
Contractual work scope of the Project and its requirements,

- Project objectives and its priorities,
- Project progress control,
- Design, procurement, construction and commissioning plan,
- Development of necessary quality management system and its improvement,
- HSE Plan.
- Required verification, validation, monitoring and quality auditing, inspection and test activities, acceptance criteria specific to the product and service,
- Quality records needed to provide evidence that the realization processes and resulting products meet requirements, etc.

The work activities shall be planned and performed serially in accordance with applicable procedures.

7.1.4 Project Schedule

The project schedule shall be formulated by the discipline groups to include all necessary work activities and work.

The project schedule shall be broken down to the total scope of work into engineering, procurement, construction and commissioning activities that will identify the controls required planning, monitoring and reporting the project progress.

All planning activities related to design and engineering will reflect the milestones for design, engineering, drawings preparation, design approvals, equipment requisitioning of equipment/materials and procurement and subsequent release for construction documents after subject discipline interface.

7.1.5 Project Coordination and Monitoring

The Project Coordination Procedure is established for the control of each project activity and smooth coordination among the related parties.

The detailed engineering and procurement activities are concentrated in the Home Office for the execution of this Project.

PM shall reside in the Home Office and shall be responsible for the coordination of the design, engineering, procurement, construction and commissioning activities.

PM, assisted by the MM, QM, EM, PPM and etc. shall closely monitor the status of the Project at all levels, to ensure that actual or potential problem areas are highlighted and addressed.

7.2 Communication with OWNER

Communications with OWNER for the following shall be handled and controlled through adequate methods.

- Review and determination of the Contract and OWNER requirement
- Change of the Contract and design condition during project period
- Handling of OWNER feedback including OWNER complaints

The monthly progress reports ("MPR") shall be prepared by PM and issued to the OWNER in accordance with Project Coordination Procedure.

The report shall inform the OWNER and management of the time-related status of the Project and shall specifically address problem areas, critical items and other areas of concern, such as inclement weather, industrial actions and late deliveries, which could adversely affect the Project.

The MPR should also indicate what actions the Contractor is taking to mitigate or resolve the concern raised.

7.3 Procurement Control

All procurement activities, purchasing and subcontracting, from Home Office and/or Site Office, shall be performed, managed, controlled, and verified in accordance with documented procedures to ensure that procured products or services conform to specified project requirements and OWNER requirements.

Procurement activities include, but are not limited to, subcontracting, purchasing of equipment and materials, and related expediting, inspection and testing, traffic control, material control, and tracking.

Procurement shall be performed in accordance with Procurement Procedure and its related procedures by following steps.

7.3.1 Control of Procurement Documents

1) Preparation of Inquiry Document

Technical Inquiry Document

On the basis of developed basic process design and engineering, relevant design EM shall prepare technical inquiry documents.

The technical inquiry documents shall comprise the following:

- General requirements of vendor's drawings, data, specifications, etc.
- Basic technical data, specifications and drawings, applicable code, special process qualification, procedures, etc.
- Inspection and test specifications, acceptance criteria, etc.
- Quality & HSE requirements, etc.

Commercial Inquiry Document

Procurement member shall prepare commercial requirements, such Instruction to bidder, terms and conditions of purchase, shipping and packing Instructions, etc.

2) Review of Inquiry Document

Technical inquiry document shall be reviewed by EM overall.

Whole Inquiry documents as well as commercial inquiry document shall be reviewed by PPM for the adequacy of specified requirements prior to release.

After the preparation of Inquiry documents, PPM shall prepare Inquiry Package in coordination with EM.

For categorization, grouping inquiries of similar equipment shall be taken into consideration.

The Inquiry Package shall be submitted to OWNER for information, review or approval in accordance with the document classification as stated in the project specification

7.3.2 Selection of Vendor

1) Vendor List

PPM shall select proposed vendors from the approved Project Vendor List basis of their ability to meet inquiry, product particulars, past experiences.

2) Additional Vendor List

Any additional vendors proposed by the Contractor shall be subject to review and approval by Owner.

3) Vendor Survey

Vendor survey shall be performed by COTRACTOR surveyor designated by PPM and OWNER may attend the vendor survey if needed. After performing the vendor survey, vendor evaluation results shall be reported to PM and relevant disciplines by COTRACTOR surveyor.

4) Review of Quotation & Vendor Evaluation

Upon receipt of quotation, PPM shall check the quotation with the cooperation of relevant procurement engineer to ensure that the quotation fully conforms to the requirements of inquiry document.

If there are errors or any omission, PPM shall request the vendor to make correction or complete the requirements.

Selection of vendor shall be totally dependent upon the type of product and service, vendor's demonstrated present capacity, vendor's quotation, quality evaluation and capability on schedule.

CONTRACTOR shall be responsible, for the quality assurance system evaluation of vendor quotations in accordance with Procurement Procedure and relevant procedures

The evaluation of vendor's quality system during execution is accompanied by vendor surveillance or shop inspection and performed by Contractor's Inspector with checklist provided from QM.

In case vendors consider it necessary, vendor may propose deviation and/or alternative list for consideration. If the proposal is beneficial to the project, the proposal may be submitted to Owner as a deviation for his approval prior to proceeding.

7.3.3 Evaluation and Verification of Vendor's Performance

After placing purchase order, CONTRACTOR exerts the following control:

1) Review and Approval of Vendor's Documents

Under the coordination of Document Controllers, vendor's documents shall be reviewed and approved by EM for their compliance to the requisition. Where it deems necessary, Document Controllers shall send them to relevant engineers for squad check.

2) Control and Verification of Vendor's Performance

A kick-off meeting will be held between CONTRACTOR and vendor soon after contract award.

Pre-inspection meeting will be held prior to fabrication and inspection, when CONTRACTOR/vendor deems it necessary.

Shop inspection shall be performed in accordance with Shop Inspection Procedure/Plan and relevant procedures to verify the compliance of products to the specification.

CONTRACTOR shall provide the OWNER with adequate notification of Kick Off Meeting Schedule. The Owner may attend kick off meeting for category A equipment items.

3) Expediting

For confirmation of vendor's schedule and expediting, monthly schedule meeting shall be held with each vendor at Home Office or, when CONTRACTOR so requires, at vendor's shop.

In this meeting, any problems related to vendor's progress schedule, inspection schedule, delivery and document issue schedule are discussed and necessary expediting shall be carried out.

7.3.4 Production Process Control

1) Product Identification and Traceability

Materials, parts and components to be identified shall be specified in specification, drawing inspection program, etc.

In accordance with the product shape, material, etc., identification shall be performed either by employing stamp, tag, label, nameplate, or by color-coding segregation. Marking shall be clear and legible and not leave detrimental effect on the product.

For in-process identification, following quarantine procedure shall be applied:

- Transfer of marking
- Identification of welding filler materials

- Identification and separation during all stages of production
- Identification and separation of non-conforming products
- Identification records and certificates of materials shall be traceable to the products and relevant documents.

2) Fabrication Process Control

CONTRACTOR shall define requirements such as, applicable codes/standards, test method, qualification, quality plan, environmental control, right to access, etc., in the technical inquiry.

Prior to the start of significant work, EM shall hold kick-off meeting between CONTRACTOR and vendor for mutual understanding.

Inspector shall supervise, monitor, as required, to ensure that the specified requirements are met.

3) Special Fabrication Process Control

Special fabrication processes such as, welding, heat treatment, nondestructive examination (NDE), special leak tests, etc. shall be performed by the procedures, equipment, personnel, and proper auxiliary materials specified or qualified in accordance with applicable codes and standards or specification designated by CONTRACTOR.

Vendor shall submit the qualification test records and/or qualified personnel list for review and approval by CONTRACTOR.

Vendor shall perform and continuously monitor the fabrication process in accordance with the approved procedure by qualified personnel or adequately proficient personnel using calibrated equipment.

CONTRACTOR Inspector shall check and monitor the process history through the vendor's inspection & test reports at any time to confirm the compliance with the requirements.

Records for special processes shall be maintained.

4) Material Identification and Traceability

It shall be the responsibility of the PPM or his representative to ensure that, during the production of equipment and systems at vendor's shop, all composite materials used are identified against the relevant specification requirements.

PMI (Positive Material Identification) Procedure for special materials shall be issued by QM.

Material identification & traceability for general materials shall be by means of documentation and certification pertaining to the materials test records. These records shall be checked as to conformity with the purchase order and relevant specification/code requirements for this project.

Vendor's inspection & test reports, including material test certificates, NDE reports, etc. shall be checked by QM or his representative.

During the construction phase of the project it shall be the responsibility of the MM to ensure that all materials delivered, prefabricated and erected are properly identified and fully in adherence/conformity with the requirements of the project specifications.

5) Handling, Storage and Shipping

- ***Planning and Controlling General***

PPM shall plan and arrange required specifications etc. on handling, storage, and shipping on the basis of regulation, OWNER requirements and CONTRACTOR Standard in the procurement documents.

Where required, vendors shall submit required specification and procedures for CONTRACTOR review and approval.

- ***Handling***

Vendors shall provide means of handling that prevent damages or deterioration.

Handling of heavy components shall be performed under controlled condition.

Operators of lifting equipment shall be qualified or well trained. Periodic inspection of lifting equipment, tools and/or jigs shall be performed, if necessary.

- ***Storage***

Products to be stored for a long period shall be prevented from misuse and deterioration by employing such measures as environmental control, identification, preventive maintenance provided by vendor.

In order to detect deterioration, conditions of products in stock shall be checked at appropriate intervals using checklist, and when applicable, in accordance with the vendor's preventive maintenance procedure.

- ***Packaging and Shipping***

PPM shall prepare shipping and packing instructions as a part of procurement document.

Prior to packaging, vendor shall inspect the cleanliness, coating, application of rust preventive means, required records of the product, material used for packaging, identification, quantity, etc.

Vendor shall perform packaging, marking and preservation and prepare shipping documents in accordance with the instruction.

Shipment shall be arranged and scheduled with the coordination of the shipping agency and/or forwarder who has been selected by CONTRACTOR.

Vendors shall notify the cargo ready date, and deliver the cargo to the specified location within the term of delivery.

7.4 Construction Control

7.4.1 Construction Planning

MM has responsibility to establish construction execution planning in accordance with OWNER requirements and relevant CONTRACTOR Standards and shall issue control procedures covering field control elements before opening the site

7.4.2 Construction Schedule Control

A Project construction schedule shall be developed and controlled in accordance with Project Control Procedure.

This construction schedule shall fully identify the applicable activities that control the manufacturer, prefabrication and installation of each item.

The construction schedule shall be reviewed and monitored by the PMM and the Project Planning/Scheduling Engineer, in order to highlight possible areas that may affect the overall Project schedule and completion

7.4.3 Field QA/QC Planning

Quality Manager has responsibility to establish construction field QA/QC execution planning in accordance with OWNER requirements and relevant CONTRACTOR Standards and shall issue control procedures covering field control elements.

7.4.4 Construction Materials Control

Materials control at construction site shall be subjected in accordance with Material Control Procedure at site.

1) Material Receiving Inspection

The QC shall be responsible for the receiving and inspection of plant materials, equipment and construction supplies purchased for this Project.

The material controller shall survey all goods received for evidence of damage and shortage.

2) Material Handling, Storage, Delivery and Use

Material controller shall be responsible for the care, custody and distribution of all material and equipment, also ensuring that the identification, coding, tagging and certification are correctly recorded.

Particular attention shall be given to preventive maintenance or other specific instructions of stored items are met, for compliance to equipment warranties, Contract requirements and vendor data packages.

Material shortage for construction shall be expected and suitable site order shall be issued with an enough handling time.

7.5 Servicing

For any servicing required by the Contract, project manager or his designee will establish and maintain documented procedures for planning, performing, controlling, verifying, and reporting that the servicing activities meet the specified requirements.

7.6 Control of Monitoring and measuring Devices

Procedures will be developed and implemented to control, calibrate and maintain monitoring and measuring devices needed to provide evidence of conformity of products to specified requirements.

Monitoring and measuring devices will be used in a manner ensures that the measurement uncertainty is known and is consistent with the required measurement capability. Where test software or test hardware are used as suitable forms of inspection, they will be checked to prove that they are capable of verifying the acceptability of products prior to release, and will be rechecked at prescribed intervals. The extent and frequency of such checks will be established and records maintained as evidence of control.

CONTRACTOR will require similar procedures and practices from its subcontractors who need calibrated monitoring and measuring devices to assure compliance to specifications for their products. Control log and calibration sheets of monitoring and measuring equipment shall be maintained.

7.7 Records

Records shall be collected and compiled by the relevant engineers with prior agreement and allotment.

Records shall include the results of analysis and evaluation, corrective actions, changes and modification, calibration records, etc.

7.8 Site HSE Management

It shall be the joint responsibility of the MM to initiate a health, environment and safety control-training program of Construction HSE Plan.

All personnel who will be employed on the construction site shall attend this course as a matter of priority. Personnel which require additional training to meet specific Contract requirements shall be subject to the training stipulations of Construction HSE Plan.

It shall be the responsibility of the MM to appoint a Site HSE Manager.

The Site HSE Manager shall prepare a HSE control instruction to cover the construction site activities.

It shall be responsibility of the Site HSE Manager to ensure that all site personnel and visitors adhere fully to these instructions by means of continual monitoring and regular auditing.

8. MEASUREMENTS, ANALYSIS AND IMPROVEMENT

8.1 General

PM shall plan and implement the monitoring, measurement, analysis and improvement processes needed to

- To demonstrate conformity of products
- To ensure conformity of the quality management system
- To continually improve effectiveness of quality management system
- To achieve quality objectives
- To rectify findings of internal quality audit and quality surveillance
- To reduce nonconformity

8.2 Monitoring and Measurement

8.2.1 Project Quality Audit

During the course of project execution, internal quality audits by QM shall be conducted at planned intervals to determine that the results of quality activities conform to the requirements of quality management system established by the project organization.

Effective implementation and maintenance of quality management system will also be audited. QM shall issue a report with documented advice, as to areas of non-conformance to this PQP.

Audit findings will be tracked to ensure proper closure and effectiveness of corrective action.

Subject to the extent of non-conformance, the QM may conduct additional audits to those identified on the audit schedule accordingly to avoid re-occurrence.

CONTRACTOR'S audit activities of vendors and sub-contractors shall be undertaken at the work place to the extent required for the scope of work being audited.

The audits for vendors and sub-contractors shall be performed by QA, if required.

8.2.2 Monitoring and Measurement of Processed

PM and MM shall apply suitable methods for monitoring and, where applicable, measurement of the work processes and performance of the Project.

Monitoring and measuring activities on quality related processes shall be carried out in accordance with the established implementation plan to achieve the project objectives and to ensure conformity of the products.

When the results of monitoring and measuring activities of processes are unsatisfactory, the managers who are responsible for the applicable matters shall timely correct and take corrective action under control of PM and/or MM and under coordination with QM, to ensure conformity of the product.

QM shall verify the results of monitoring and measurement of processes and suitability of action taken against unsatisfactory results and its effectiveness.

8.2.3 Monitoring and Measurement of Product

1) Shop Inspection and Test Program

On the basis of equipment specification prepared by QM shall develop Shop Inspection Plan and relevant Procedure in which applicable code/standard, items to be inspected, kinds and level of inspection, witness inspection points, etc. are identified.

Vendors are requested establish detailed Inspection & Test Procedure/Plan, in which detailed inspection and tests methods, fabrication sequences, measuring equipment, acceptance criteria, required records, etc. shall be defined.

When inspection is impossible or disadvantageous, selective control by monitoring the process will be employed.

Inspection Execution Control

Pre-Inspection Meeting shall be held for major products between CONTRACTOR and vendor for coordination and full understanding prior to fabrication and inspection.

Vendor shall establish inspection schedule on the basis of delivery date. Vendors shall notify inspection date and inspection status at specified interval to QM for his control.

At shop inspection, inspector shall check applicable vendor drawings and inspection and/or fabrication procedure have been properly approved before commencing inspection.

Witness inspection will be conducted at vendor shop by the inspector at the witness point. OWNER representative may attend the inspection when he deems it necessary.

Inspection Record

On completion of final shop inspection, inspector shall check vendor's inspection & test record and reports. The inspection report shall be submitted to OWNER with the inspection release certificate (IRC).

2) **Field Inspection and Testing**

Inspection and test at site required to ensure that specified requirements for the product are met shall be controlled by written procedures including inspection and test plans.

All required notifications of inspections are sent to OWNER and/or their authorized inspectors in a timely manner to allow their participation therein.

Construction SI shall develop project-specific ITPs under discussion with QC Those ITPs shall cover project inspection and test requirements for construction products, and will be submitted to OWNER for approval, if required.

Receiving Inspection

Field QC Procedure shall ensure that incoming equipment and materials procured shall not be used or processed (except in the circumstances described below) until it has been successfully inspected or otherwise verified as conforming to specified requirements. Verification of the specified requirements shall be in accordance with the inspection and test plans and/or documented procedures. Records of receiving inspection shall be maintained.

In determining the amount and nature of receiving inspection, consideration shall be given to the amount of control exercised at the vendor's premises and the recorded evidence of conformance provided.

When an incoming product is released for urgent use prior to verification, it shall be positively identified and recorded in order to permit immediate recall and replacement in the event of nonconformity to specified requirements.

In-Process Inspection and Test

When in-process inspection and test for the construction product is required, the procedures shall require hold-points, i.e., holding of product from further processing until the required inspection and tests have been completed or necessary reports have been received and verified.

Final Inspection and Test

When required, final inspection and test shall be carried out in accordance with a prepared ITP and/or documented procedure to complete the evidence of conformance of the finished product(s) to specified requirements.

Inspection and test plans and/or documented procedures for final inspection and testing shall require that all specified inspection and tests, including those specified either on receipt of product or in-process, have been carried out and that the results meet specified requirements.

No product shall be dispatched or used until all the activities specified in the ITP and/or documented procedure have been satisfactorily completed and the associated data and documentation are available and authorized.

8.2.4 Control of Non-Conformances

It shall be the responsibility of the relevant LE or his representative to institute corrective actions for reported non-conformance.

Reporting of non-conformance which are not identified during internal audit shall be the responsibility of the relevant LE. Non-conformities in design/engineering works will be reported, handled by EM.

Non-conforming materials and products purchased by CONTRACTOR shall be addressed in accordance with Shop Inspection Procedure/Plan for the purposes of this Project.

Non-conformities in installation/erection of equipment and materials will be reported, handled in accordance with Field Coordination Procedure and Field QC Procedure.

Any request for deviation or waiver to OWNER, for materials or products, resulting from non-conformance advice, shall be subject to written approval of OWNER.

8.2.5 Corrective and Preventive Actions

Project organization will function proactively on the project activities to prevent quality-related problems. Nonconformity may be identified through activities such as inspections, audits, surveillance, trend analyses, owner complaints, etc.

If nonconformity is noted, it shall be promptly identified and appropriate corrective, and where necessary, preventive action shall be initiated in a timely manner.

Corrective and preventive action taken will correct any nonconformity, and project organization shall strive to eliminate the root cause of the identified or potential nonconformity.

Deviations from approved designs and/or approved quality system procedures noted during quality audits or quality surveillance will be documented using audit finding report, corrective action or preventive action report form.

Procedure for corrective and preventive action shall be established including the following;

- Review of identified or potential nonconformity,
- Determination of the cause of identified or potential nonconformity,
- Evaluation of the need for action to prevent recurrence or occurrence of nonconformity,
- Determination of action needed and its implementation,
- Record control,
- Follow-up on the actions taken.

All identified and potential nonconformity are reviewed, evaluated, disposed, and approved by authorized persons.

8.2.6 Continual Improvement

Project organization shall review and use the results of data analysis to improve continually the effectiveness of quality management system and the applicable project work.

Continual improvement shall be implemented according to improvement plan. When necessary, improvement plan shall be revised according to achievement level of improvement and situation changes.

Project organization shall continually confirm and evaluate progress status, results of improvement activities during project period. If progress and effectiveness is unsatisfactory, improvement plan shall be revised and countermeasure shall also be established and implemented.

8.2.7 Quality Record

1) Quality Records

Documented quality audits and records of reviews, which verify that the activities related to design, engineering, procurement (including inspection and testing) and construction are being performed in accordance with the requirements of this PQP and Project Coordination Procedure, shall be generated progressively and filled in a centralized location for ease of retrieval and reference.

These records shall include all related instruction and actions issued during the course of the project to ensure that the quality of the work is maintained at all times.

These records shall include, but not limited to, the following:

- Quality control records/reports
- Audit/corrective action reports
- Non-conformance reports in design, engineering, procurement, construction, and commissioning phase
- Deviation disposition request
- Assessment of Vendor, Sub-contractor
- Subcontractor/Vendor Performance Records, etc.

2) Review on Quality Records

All records related to quality shall be reviewed and endorsed on an ongoing basis by the QM or his representative.

These records shall be available at all time for evaluation and review on the Project.

When the Project shall be completed, PM or his representative shall review and check all quality records related to engineering, procurement, construction and commissioning activities, which will be included in the project data package forming the final construction field report and project close-out documentation.